sPHENIX Internal Note

Eta Edge Migration Study: Fake Rate from Fiducial Boundary Effects

Isolated Photon Cross-Section in p+p at $\sqrt{s} = 200$ GeV

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Abstract

This note documents a study of pseudorapidity edge migration effects in the sPHENIX isolated photon cross-section measurement (PPG12). Both the truth and reconstructed η fiducial cuts are set at $|\eta| < 0.7$, which allows truth photons produced outside the fiducial volume to have reconstructed clusters migrate inside, and vice versa. In the unfolding framework, inward-migrated clusters matched to truth photons outside $|\eta| < 0.7$ should be treated as fakes but currently enter the response matrix as signal in `RecoEffCalculator_TTreeReader.C`. We quantify this contamination for both single-interaction and double-interaction MC, finding a signal region (ABCD region A) contamination of ∼1.2% for single interaction and ∼0.3% for double interaction, with significantly larger effects in the non-tight ABCD regions used for background estimation.

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1 Introduction

The PPG12 analysis measures the isolated prompt photon production cross-section in p+p collisions at $\sqrt{s} = 200$ GeV using the sPHENIX EMCal. The fiducial pseudorapidity acceptance is $|\eta| < 0.7$, corresponding to the barrel EMCal coverage. This cut is applied identically to both truth-level photons and reconstructed clusters.

Because the η boundary is a sharp geometric cut rather than a smooth efficiency edge, photons near $|\eta| \approx 0.7$ can migrate across the boundary between truth and reconstruction:

- **Inward migration (fakes):** Truth photons with $|\eta^{\text{truth}}| > 0.7$ produce reconstructed clusters with $|\eta^{\text{reco}}| < 0.7$. These enter the signal sample but have no corresponding truth entry within the fiducial volume.
- **Outward migration (losses):** Truth photons with $|\eta^{\text{truth}}| < 0.7$ produce reconstructed clusters with $|\eta^{\text{reco}}| > 0.7$, escaping the reconstructed sample.

In the current implementation of `RecoEffCalculator_TTreeReader.C`, inward-migrated clusters enter the response matrix as signal — i.e., they are treated as correctly matched truth-reco pairs. This is incorrect: their truth p_{T} falls outside the fiducial region, biasing the unfolding. This study quantifies the magnitude of this effect.

2 Method

A dedicated macro `EtaMigrationStudy.C` processes the full photon MC samples: `photon5` (truth p_{T} range 0–14 GeV), `photon10` (14–30 GeV), and `photon20` (30+ GeV). Each sample is weighted by its production cross-section.

Truth photons are registered with *no* η filter — all direct/fragmentation photons satisfying $E_{\text{T}}^{\text{iso,truth}} < 4$ GeV (cone $R_{\text{cone}} = 0.3$) are included regardless of η. Each truth–reco matched pair is then classified into four categories based on the fiducial boundary $|\eta| < 0.7$:

1. **Good:** both truth and reco inside $|\eta| < 0.7$.
2. **Fake (inward):** truth outside, reco inside.
3. **Lost (outward):** truth inside, reco outside.
4. **Both outside:** neither truth nor reco in fiducial volume.

The standard PPG12 ABCD classification is applied to each pair:

- Region A (tight + isolated): signal region.
- Region B (tight + non-isolated): background template.
- Region C (non-tight + isolated): background control.
- Region D (non-tight + non-isolated): background normalization.

For the double-interaction study, a toy vertex shift is applied: a random second vertex is drawn from the data vertex distribution, and the new vertex is computed as $z_{\text{vtx}}^{\text{new}} = (z_{\text{vtx}}^{\text{orig}} + z_{\text{vtx}}^{\text{rand}})/2$. Cluster kinematics are recalculated using the CEMC cylinder geometry ($R = 93.5$ cm):

$$z/r_{\text{new}} = z/r_{\text{old}} + \frac{z_{\text{vtx}}^{\text{old}} - z_{\text{vtx}}^{\text{new}}}{93.5\ \text{cm}}, \quad \eta_{\text{new}} = \sinh^{-1}(z/r_{\text{new}}), \quad E_{\text{T}}^{\text{new}} = E/\cosh(\eta_{\text{new}}). \tag{1}$$

3 Results

3.1 Overall Migration Rates

Table 1 summarizes the unweighted migration rates per MC sample. The fake (inward) rate is defined as the number of inward-migrated pairs divided by the total number of matched pairs, and similarly for the loss (outward) rate.

Table 1: Overall migration rates (unweighted) for single-interaction MC. Fake rate = inward / total; loss rate = outward / total.

Sample	Total	Good	Fakes	Losses	Both out	Fake rate	Loss rate
photon5	2361432	1523346	18490	13687	805909	1.20%	0.89%
photon10	532878	374380	3942	5822	148734	1.04%	1.53%
photon20	128595	107741	721	2193	17940	0.66%	1.99%

The fake rate decreases with increasing truth p_{T} range (from 1.20% to 0.66%), while the loss rate increases (from 0.89% to 1.99%). The asymmetry arises because higher-p_{T} photons have narrower shower profiles and more precise η reconstruction, but the steeply falling spectrum preferentially populates the inward direction at lower p_{T}.

3.2 Signal Region (A) Contamination

The physically relevant metric is the contamination of the ABCD signal region (region A: tight BDT + isolated). This is defined as:

$$\text{Contamination} = \frac{N_{\text{fakes}}^{\text{tight+iso}}}{N_{\text{good}}^{\text{tight+iso}} + N_{\text{fakes}}^{\text{tight+iso}}}. \tag{2}$$

This quantity directly measures the fraction of response matrix entries that are edge-migration fakes.

3.2.1 Single Interaction

Table 2 shows the signal region contamination per p_{T} bin for single-interaction MC.

The contamination is approximately flat at ∼1.2–1.3% across most of the p_{T} range but rises to 2.57% in the highest bin (32–36 GeV).

3.2.2 Double Interaction

Table 3 shows the equivalent for double-interaction (toy vertex shift).

The double-interaction contamination in region A is significantly *lower* than single interaction (0.34% vs. 1.24%). This counterintuitive result is explained in Section 3.4.

3.3 ABCD Region Breakdown

The edge migration does not affect all ABCD regions equally. The contamination varies strongly by ABCD region, particularly in the highest p_{T} bins where the effect is most pronounced.

Region C (non-tight + isolated) exhibits dramatically higher contamination, reaching up to 52% at 32–36 GeV for single interaction. This occurs because edge-migrated clusters have distorted

Table 2: Signal region (tight + isolated) contamination from η edge migration, single-interaction MC. Counts are cross-section weighted.

p_{T} bin (GeV)	Good (tight+iso)	Fakes (tight+iso)	Contamination
8–10	1.04e8	1.30e6	1.24%
10–12	3.71e7	4.43e5	1.18%
12–14	1.45e7	2.06e5	1.40%
14–16	6.28e6	7.87e4	1.24%
16–18	2.86e6	3.76e4	1.30%
18–20	1.35e6	1.79e4	1.31%
20–22	6.62e5	8.68e3	1.29%
22–24	3.41e5	4.42e3	1.28%
24–26	1.79e5	2.34e3	1.30%
26–28	9.19e4	1.61e3	1.72%
28–32	7.70e4	1.49e3	1.90%
32–36	2.11e4	5.57e2	2.57%
Overall	1.67e8	2.10e6	**1.24%**

Table 3: Signal region (tight + isolated) contamination from η edge migration, double-interaction MC (toy vertex shift).

p_{T} bin (GeV)	Good (tight+iso)	Fakes (tight+iso)	Contamination
8–10	7.81e6	2.58e4	0.33%
10–12	2.78e6	1.12e4	0.40%
12–14	1.00e6	3.37e3	0.33%
14–16	4.09e5	1.33e3	0.33%
16–18	1.79e5	3.19e2	0.18%
18–20	8.37e4	2.66e2	0.32%
20–22	4.00e4	2.13e2	0.53%
22–24	2.07e4	1.07e2	0.51%
24–26	1.10e4	0	0.00%
26–28	5.82e3	53	0.91%
28–32	4.87e3	1	0.02%
32–36	1.39e3	1	0.07%
Overall	1.24e7	4.27e4	**0.34%**

shower shapes: the η mismatch between truth and reco positions distorts the transverse shower profile, causing these clusters to preferentially fail the tight BDT selection and land in the non-tight category.

The signal region A contamination remains modest (1.2–2.6%) precisely because the BDT effectively rejects edge fakes — they carry anomalous shower shapes that the BDT was trained to discriminate against.

3.4 Double Interaction Enhancement

The overall *raw* inward fake rate (before ABCD classification) is strongly amplified by double interaction: 1.2% (single) vs. 13.7% (double), approximately an 11× amplification. The vertex displacement in double-interaction events shifts the reconstructed η of all clusters, dramatically increasing the probability of boundary crossing.

However, the signal region contamination moves in the *opposite* direction: 1.24% (single) → 0.34% (double). The explanation is that vertex-shifted clusters acquire additional shower shape distortion on top of the edge migration effect. The compounded distortion causes the tight BDT to reject these clusters even more aggressively, suppressing their entry into region A.

Figure 9 quantifies this by showing the fraction of ABCD-classified fakes (i.e., those passing common quality cuts) that leak into regions A and C, as a function of E_{T}. For single interaction, ∼50% of ABCD-classified fakes pass tight+iso (region A) and ∼40% land in region C. For double interaction, a *higher* fraction (∼70–85%) passes tight+iso — a seemingly counterintuitive result. The explanation is a *selection effect*: the common quality cuts reject 98.4% of double-interaction inward fakes (vs. 81% for single), acting as a much stronger pre-filter. The 1.6% of double fakes that survive are the "best" fakes — truth photons barely outside the boundary with small vertex shifts — and these naturally also tend to pass the tight BDT. The common cuts are doing the heavy lifting; their survivors are biased toward photon-like signatures. The absolute signal region contamination (Fig. 7 vs. Fig. 8) remains much lower for double interaction (0.34% vs. 1.24%) because the total number of ABCD-classified fakes is ∼60× smaller.

It should be noted that the double-interaction statistics are marginal: only 77 unweighted tight+iso fake entries contribute to the entire double-interaction signal region, limiting the precision of the contamination estimate.

3.5 Response Matrix Contamination

The response matrix in `RecoEffCalculator_TTreeReader.C` is filled for all tight+iso matched pairs. The 1.2% signal region contamination from Table 2 directly translates to 1.2% of response matrix entries having incorrect truth p_{T} (from outside the fiducial η region). These entries map reco p_{T} values to truth p_{T} values that should not participate in the unfolding, introducing a bias in the unfolded cross-section.

The effect is p_{T}-dependent: rising from ∼1.2% at low p_{T} to ∼2.6% at 32–36 GeV. This p_{T} dependence means the bias is not a simple overall normalization shift but distorts the shape of the unfolded spectrum.

3.6 ABCD Leakage Fractions: Single vs. Double Interaction

The ABCD signal leakage fractions c_B, c_C, c_D (defined as the ratio of truth-matched signal in regions B, C, D to that in region A) directly enter the background subtraction and purity correction. Figure 12 shows these fractions computed from the "good" signal (truth inside fiducial $|\eta| < 0.7$) for both single and double interaction.

- c_B **(tight, non-isolated):** Increases from ∼6–11% (single) to ∼11–29% (double), a factor ∼2–3×. The vertex shift in double-interaction events smears the topo-cluster isolation sum, causing isolated photons to fail the isolation cut and migrate from region A to B.

- c_C **(non-tight, isolated):** Comparable between single and double (∼2–8%), with no systematic amplification. The shower shape BDT is less sensitive to vertex shifts at the level relevant here.

- c_D **(non-tight, non-isolated):** Small in both cases ($< 2\%$), with modest increase for double interaction.

The dominant effect is the c_B amplification. Since the ABCD background estimate is $N^A_{\text{bkg}} = R \cdot B \cdot C/D$ with signal leakage corrections proportional to $(1 - c_B)(1 - c_C)/(1 - c_D)$, the 2–3× increase in c_B for double interaction significantly affects the corrected purity, particularly at high E_{T} where the leakage is largest.

4 Discussion

The ∼1.2% signal region contamination is a systematic effect, not a statistical fluctuation. At the highest p_{T} bin (32–36 GeV), the contamination reaches 2.6%, which is comparable in magnitude to other systematic uncertainties in the analysis (e.g., energy scale, BDT threshold variations).

Three approaches could address this:

1. **Asymmetric η cuts:** Apply a tighter reco η cut than truth (e.g., $|\eta^{\text{reco}}| < 0.6$ with $|\eta^{\text{truth}}| < 0.7$). This eliminates inward migration but sacrifices acceptance.

2. **Truth η check in response matrix:** Explicitly verify that the truth η of matched pairs satisfies $|\eta^{\text{truth}}| < 0.7$ before filling the response matrix. Outside-fiducial matches are counted as fakes rather than signal. This is the minimal fix.

3. **Systematic uncertainty:** Include the contamination as a systematic uncertainty on the unfolded cross-section, propagated through the full pipeline.

The region C contamination (up to 52% at high p_{T}) is a separate concern. Region C enters the ABCD background estimation through the R-factor: $R = N_C/N_D$. If a large fraction of region C entries are edge-migration fakes rather than genuine non-tight background, the R-factor and consequently the background subtraction could be biased.

For double interaction, the tight+iso selection provides effective rejection of edge-migrated clusters (contamination drops from 13.7% raw to 0.34% in region A). However, the ∼14% raw migration rate is large enough to significantly affect the non-tight regions used in background estimation.

5 Recommendations

1. **Short term:** Add the truth η check to the response matrix filling in `RecoEffCalculator_TTreeReader.C`. Matched pairs with $|\eta^{\text{truth}}| > 0.7$ should be counted as fakes, not signal.

2. **Medium term:** Evaluate the impact on the unfolded cross-section by running the full pipeline with and without the fix. The difference quantifies the bias introduced by the current implementation.

3. **Background estimation:** Evaluate and include the region C contamination as a systematic uncertainty on the ABCD R-factor.

6 Appendix: Plots

Figures 1–11 show the distributions produced by `plot_eta_migration.C`.

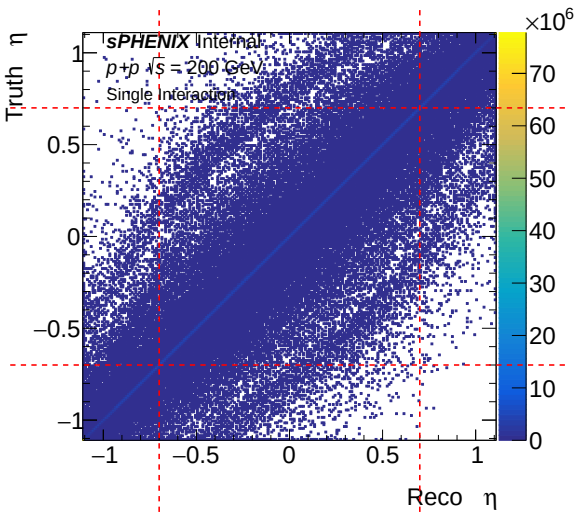

Figure 1: Migration matrix: truth vs. reco η classification for single-interaction MC.

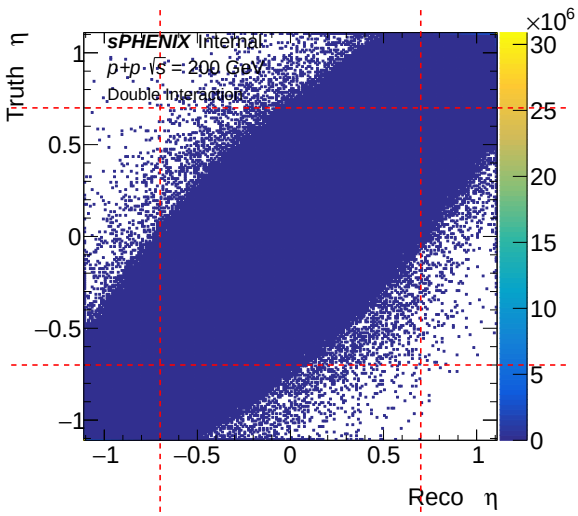

Figure 2: Migration matrix: truth vs. reco η classification for double-interaction MC (toy vertex shift).

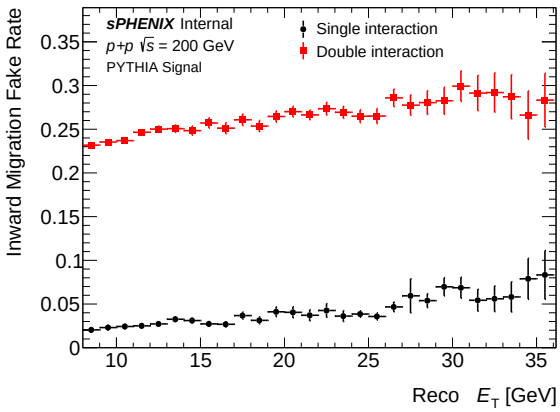

Figure 3: Inward fake rate vs. E_{T} for single and double interaction.

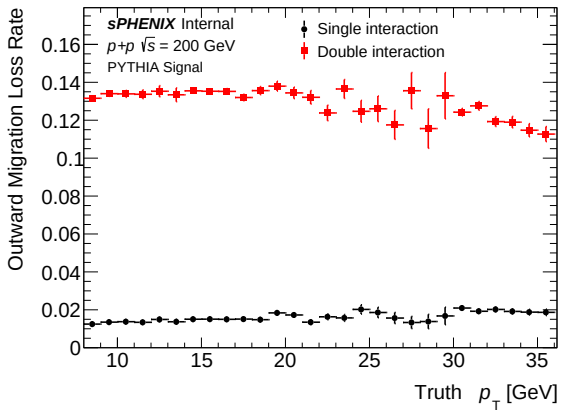

Figure 4: Outward loss rate vs. E_{T} for single and double interaction.

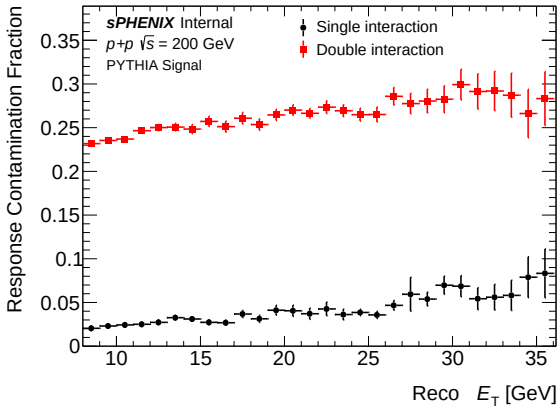

Figure 5: All-cluster inward migration fraction vs. E_{T} for single (∼2–8%) and double (∼25%) interaction. This is the raw rate before any ABCD selection.

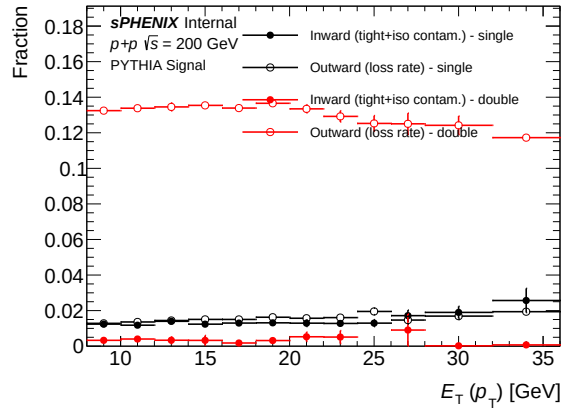

Figure 6: Response matrix impact: tight+iso inward contamination (filled circles) and outward loss rate (open squares) vs. E_{T}. Single interaction: ∼1.2% contamination and ∼1.5% loss. Double: ∼0.3% contamination but ∼13% loss.

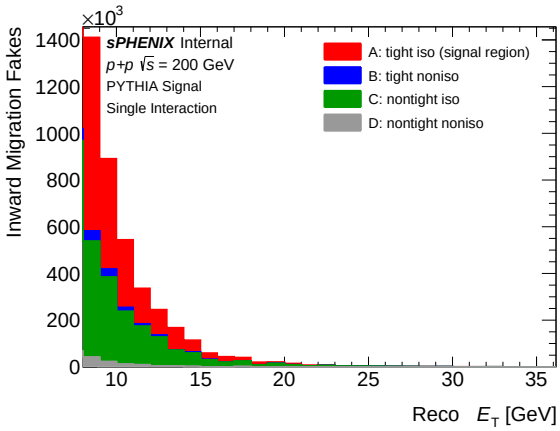

Figure 7: ABCD distribution of inward-migration fakes for single interaction. Region A (tight+iso) and C (nontight+iso) dominate.

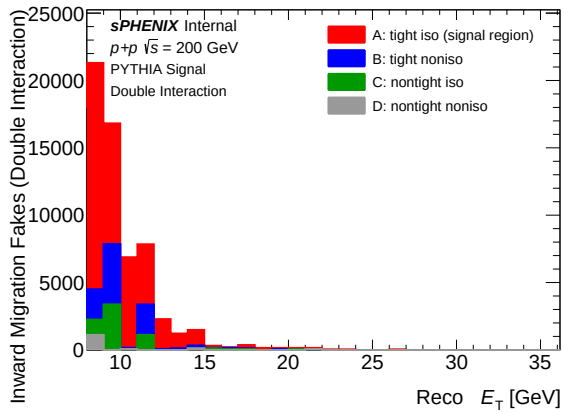

Figure 8: ABCD distribution of inward-migration fakes for double interaction. The total yield is ∼60× smaller than single interaction (Fig. 7), because 98% of double-interaction fakes fail common quality cuts.

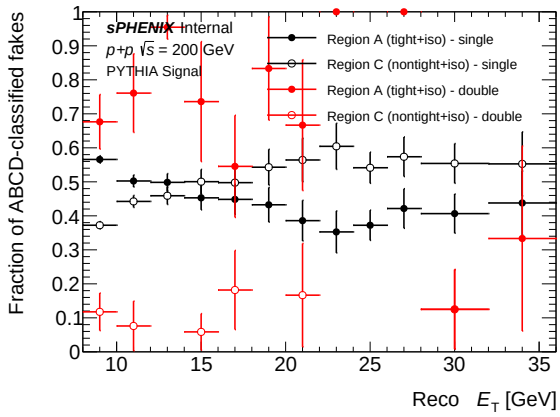

Figure 9: Fraction of inward-migrating fakes that leak into regions A (tight+iso, filled circles) and C (non-tight+iso, open squares) as a function of E_{T}, comparing single (black) and double (red) interaction. Single-interaction fakes have a higher probability of passing tight+iso, while double-interaction fakes are more efficiently rejected into region C.

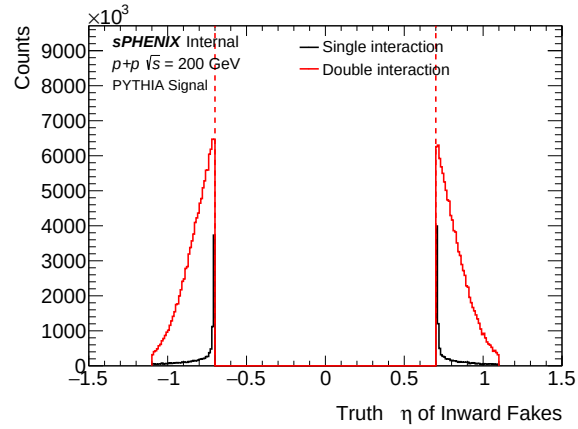

Figure 10: Truth η distribution of inward-migrated fakes.

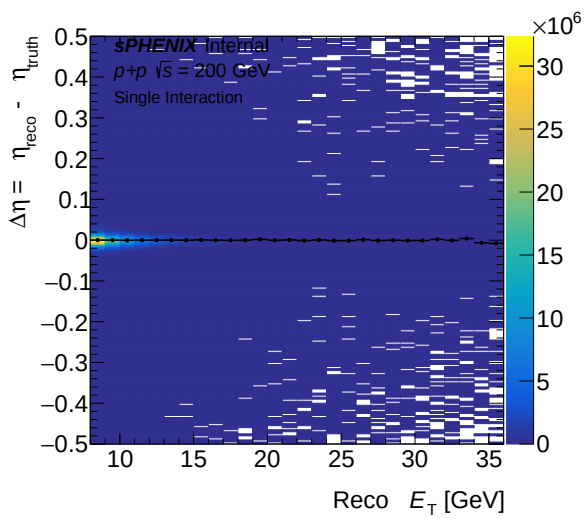

Figure 11: $\Delta\eta$ (truth $-$ reco) vs. E_{T} for edge-migrated clusters.

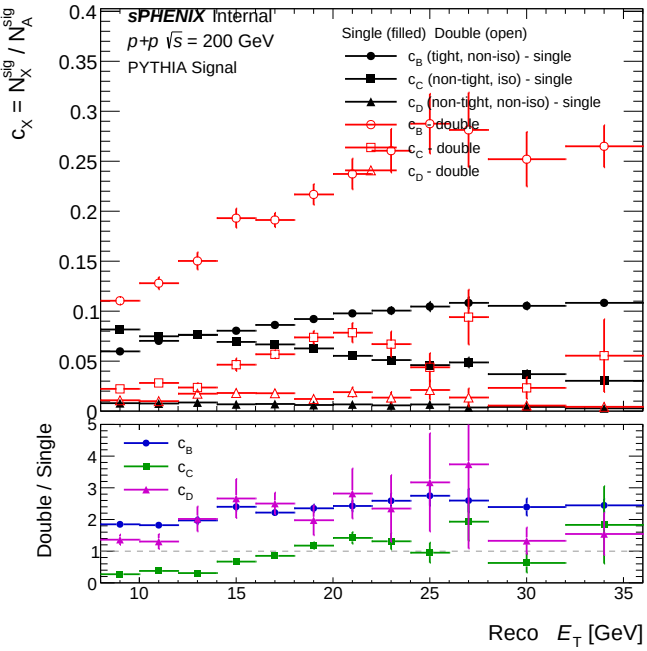

Figure 12: ABCD signal leakage fractions $c_X = N_X^{\mathrm{sig}}/N_A^{\mathrm{sig}}$ for single interaction (filled markers) and double interaction (open markers) as a function of E_{T}. Upper panel: absolute leakage fractions for regions B, C, and D. Lower panel: double-to-single ratio. The isolation leakage c_B increases by a factor of $\sim$2–3$\times$ under double interaction due to vertex-shift smearing of the isolation cone, while the shower-shape leakage c_C is comparable or slightly reduced.